EE2002-Tutorial 1 Solution Key

1. Convert $(28.8)_{10}$ to binary. Show the result up to five places after the binary point.

Solution:

$$(28.8)_{10} = (11100.11001\ldots)_2.$$

2. Determine the positive value of x if $(211)_x = (152)_8$

Solution:

$$x = 7.$$

3. Convert $(1001.110)_2$ to Decimal. Assume input is unsigned

Solution:

$$(1001.110)_2 = (9.75)_{10}.$$

4. Convert $(0.528)_{10}$ to Octal.

Solution:

$$(0.528)_{10} = (0.4162)_8.$$

5. Calculate the sum of 1010.011 and 1110.001. Assume the numbers are in 2's complement format

Solution:

$$1010.011 + 1110.001 = (1000.1)_2 = (-7.5)_{10}.$$

6. Convert $(7562.45)_{10}$ to Hexadecimal.

Solution:

$$(7562.45)_{10} = (1D8A.73)_{16}.$$

7. Let $X = 1010100$ and $Y = 1000011$. Calculate $X - Y$ and $Y - X$. Assume the input numbers are in 2's complement format.

Solution:

$$X - Y = 17_{10}, \quad Y - X = -17_{10}.$$

8. Convert $(640.7)_8$ to Decimal.

Solution:

$$(640.7)_8 = (416.875)_{10}.$$

9. Perform $(+42) + (-13)$ and $(-42) - (-13)$ using binary arithmetic. Assume the input numbers are represented in two's complement format.

Solution:

$$(+42) + (-13) = 29_{10}, \quad (-42) - (-13) = -29_{10}.$$

10. Represent (-30.5) in Two's Complement Form

Solution:

$$(-30.5) = (100001.1)_2.$$